James Conlon

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github.com/proconlon

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Education

Boston University Boston, MA
B.S. in Computer Engineering, Minor in Computer Science (GPA: 3.53) *Sep 2022 – May 2026*

Experience

Embedded Software and Security Intern Jun 2025 – Present
ei^3 : Industrial Internet Intelligence *Pearl River, NY*

- Designed and implemented a novel OpenVPN-based TAP server solution using Linux network namespaces, policy routing, and multiple "virtual proxy" VPN clients to resolve client-side NAT and subnet overlap issues for legacy industrial devices.
- Engineered a 4-stage Python CLI and GitLab CI/CD pipeline to automate device commissioning, utilizing Golden Master testing on physical hardware to verify 12+ real-world network scenarios (Proxies, Static/DHCP, and IP Routing).
- Collaborated with internal support teams to author the SOP for automated remote deployments, establishing a safe migration process that preserves complex customer network topologies like VLANs and static routes.
- Designed and owned the REST API and data model (Python/Flask) for managing device configurations and firmware, allowing future expansion for new device models or features.

Embedded Software and Security Intern Jun 2024 – Jun 2025
ei^3 : Industrial Internet Intelligence (FT summer, PT during semesters) *Pearl River, NY*

- Engineered a configurable OpenVPN TUN server in Docker for production edge hardware; configured systemd services and Linux boot sequences to ensure persistent container orchestration and reliable hardware startup.
- Developed a multi-instance OpenVPN solution in Docker for production use on factory hardware, enabling advanced network routing (fwmark, policy routing, NAT) to secure legacy industrial devices in production.

IT Support Specialist Aug 2022 – Dec 2024
BU Information Services and Technology *Boston, MA*

- Diagnosed and resolved hardware/software issues: Windows/Mac/Linux, networking, and academic software.

Projects

Autonomous Arctic Monitoring System | *Python, RPi 5, Arduino, System Integration* Sep 2025 – May 2026

- Designing system architecture for a ruggedized, dual-node embedded camera platform intended for long-duration autonomous deployment on Arctic sea ice.
- Automating embedded Linux workflow using custom Bash scripts and systemd services to orchestrate boot sequences, manage peripheral initialization, and ensure reliable headless execution.
- Prototyping a "Sentry/Worker" topology utilizing an Arduino Nano for continuous low-power environmental monitoring to wake a Raspberry Pi 5 only for high-bandwidth tasks.
- Implemented sensor fusion logic to integrate PIR and mmWave radar data, creating a multi-stage trigger logic designed to filter environmental noise and detect wildlife.

Predictive Maintenance Edge Data Logger | *C, Python, AWS S3, Tailscale, BeagleBone* Spring 2025

- Developed an end-to-end connected edge device data logger on a BeagleBone Black, performing low-latency data acquisition from an OPC-UA server while utilizing Python for ML and cloud integration.
- Designed a dual-rate logging system to minimize cloud costs, capturing high-frequency data locally for ML model training and low-frequency data for long-term storage in AWS S3.
- Implemented secure remote access using Tailscale, air-gapping a factory network while enabling remote device management and data retrieval.
- Deployed a local Python ML model for predictive maintenance, analyzing high-frequency data to forecast equipment failures and automatically send email alerts to operators.

Technical Skills

Languages: Python, Bash, Go, C/C++
Framework & Tools: Pydantic, Flask, Linux (Debian/Alpine/RHEL/Fedora), OpenSSL
DevOps & Virtualization: Docker, Docker Compose, Git, CI/CD, Jira
Networking and Security: TCP/IP, OpenVPN, NAT, VLANs, Policy Routing, PKI, Wireshark
Courses: Systems: (Distributed, Embedded, Operating), Cybersecurity, Client-Server Software, Intro Databases